

Corrections and Clarifications

FFGFT / T0 Time-Mass Duality

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Contents

1 Correction 1: Prefactor in the Higgs Derivation of ξ	3
2 Correction 2: Tau Yukawa Prefactor r_τ	4
3 Correction 3: Base Formula for the Electron g^{-2}	5
4 Clarification 1: Precision of the Koide Formula	6
5 Clarification 2: h from the Higgs Field - Derivation Chain	6
6 Clarification 3: Two ξ Parameters - Two Different Spaces	7
7 Clarification 4: L_0 and the Schwarzschild Interpretation	8
8 Clarification 5: Scope and Claim of the Mass Formulas	9

9 Clarification 6: Koide Formula Only with Original Values

10

Preliminary Remark

This document lists corrections and clarifications to already published documents of the FFGFT series. Older documents are *not* modified. Where a contradiction exists between an older and a newer document, the version recorded here is binding.

Each entry contains: the affected document and location, the erroneous expression, the correct expression, and a brief justification.

1 Correction 1: Prefactor in the Higgs Derivation of ξ

Affected document

Doc. 049 (T0 Lagrangian and Higgs integration), HTML intermediate version and older working versions.

Erroneous expression

$$\xi = \frac{\lambda_h^2 v^2}{64\pi^4 m_h^2} \approx 1.0 \times 10^{-5} \quad (\text{old - incorrect})$$

Correct expression

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.33 \times 10^{-4} \quad (190.1)$$

with the experimental inputs:

$$m_h = 125.25 \text{ GeV} \quad (\text{Higgs mass}),$$

$$v = 246.22 \text{ GeV} \quad (\text{vacuum expectation value}),$$

$$\lambda_h = \frac{m_h^2}{2v^2} \approx 0.129 \quad (\text{Higgs self-coupling}).$$

Justification

The prefactor $64\pi^4$ originated from a typographical error in the early HTML visualisation. It yields $\xi \approx 10^{-5}$, which deviates by one order of magnitude from the geomet-

rically derived value $\xi = 4/3 \cdot 10^{-4}$ and does not reproduce the natural constants. The correct prefactor $16\pi^3$ gives $\xi \approx 1.33 \times 10^{-4}$, consistent with the independently determined value from the proton-to-electron mass ratio (Doc. 009).

2 Correction 2: Tau Yukawa Prefactor r_τ

Affected document

Doc. 116 (Koide formula as a consequence of T0 theory), table of Yukawa coefficients.

Erroneous expression

$$r_\tau = \frac{8}{3} \approx 2.667 \Rightarrow m_\tau \approx 1712 \text{ MeV} \quad (-3.6\% \text{ dev.})$$

(old - incorrect)

Correct expression

$$\boxed{r_\tau = \frac{25}{9} \approx 2.778} \Rightarrow m_\tau \approx 1783 \text{ MeV} \quad (+0.4\% \text{ dev.})$$

(190.2)

The lepton masses follow in general from:

$$m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v \quad (190.3)$$

with exponents $p_e = 3/2$, $p_\mu = 1$, $p_\tau = 2/3$ (step size $\Delta p = 1/3$) and the vacuum expectation value v .

Justification

$r_\tau = 8/3$ was a remnant from an earlier iteration stage. Doc. 158 (Koide-to-g-2 bridge) already uses $r_\tau = 25/9$, which agrees numerically with the tau mass. Furthermore, $25/9 = (5/3)^2$ has a geometric interpretation as the square of the pure sixth in the harmonic series, consistent with the musical resonance analogy (Doc. 060).

Complete corrected prefactor table

Lepton	Exponent p	Prefactor r	m [MeV], calculated
Electron	3/2	4/3	0.511
Muon	1	16/5	105.7
Tau	2/3	25/9	1783

3 Correction 3: Base Formula for the Electron $g-2$

Affected document

T0 Explorer (HTML visualisation), first version; older intermediate versions of Doc. 018.

Erroneous expression

$$a_e = \frac{4\pi}{f \cdot k_{\text{geom}}} \quad (\text{old - incorrect})$$

Correct expression

$$a_e = \frac{S_3}{f \cdot k_{\text{geom}}} = \frac{2\pi^2}{f \cdot k_{\text{geom}}} \quad (190.4)$$

with $f = 1/\xi \approx 7500$, $k_{\text{geom}} = (2/\sqrt{\varphi}) \cdot \sqrt{2} \approx 2.224$ and $S_3 = 2\pi^2 \approx 19.74$ (surface of the 4D unit sphere, i.e. the 3D boundary of the torus).

Justification

The earlier prefactor 4π corresponds to the surface of the 2D sphere in 3D space, not the geometrically correct 3D surface of the 4D torus. $S_3 = 2\pi^2$ is the correct value from the torus geometry. The fractal corrections for muon and tau continue to use 4π :

$$\Delta a_\mu = \frac{4\pi}{f^{5/3}}, \quad (190.5)$$

$$\Delta a_\tau = \frac{4\pi}{f^{4/3}}. \quad (190.6)$$

4 Clarification 1: Precision of the Koide Formula

Affected documents

Doc. 116 (Abstract, l. 14): $\Delta Q < 0.00003\%$

Doc. 158 (Abstract): “experimentally confirmed to 0.001 %”

Correct classification

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \quad (190.7)$$

is experimentally confirmed to better than 0.001 % (PDG values 2022). The statement $< 0.00003\%$ in Doc. 116 refers to the internal consistency of the T0 formulas, not to the experimental measurement uncertainty. Both statements are factually correct but describe different quantities. In external communications, 0.001 % is the appropriate figure.

Note

The individual T0 masses agree with experiment to $\sim 1\%$; the high precision of $Q = 2/3$ arises from the specific algebraic structure of the Koide combination, not from individual mass accuracy.

5 Clarification 2: $\hbar$ from the Higgs Field - Derivation Chain

Context

Planck’s constant $\hbar$ is derived differently in various documents. The complete, consistent derivation chain is recorded here.

Binding derivation chain

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \longrightarrow L_0 = \xi \cdot \ell_P \longrightarrow \hbar \sim \sqrt{\xi} \cdot \ell_P \cdot c \quad (190.8)$$

The time-mass binding condition provides the physical justification:

$$T(x, t) \cdot m(x, t) = 1 \implies \tilde{T} = \frac{\hbar}{m c^2} \implies E = \hbar \omega. \quad (190.9)$$

$\hbar = 2\pi\hbar$ is thus *not* a free constant, but a geometric consequence of ξ and ℓ_P . The numerical SI value $\hbar \approx 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ is the calibration result of this relationship to the SI base units.

6 Clarification 3: Two ξ Parameters - Two Different Spaces

Context

In documents 173–176 (quantum dots, spin qubits, qubit state spaces, Shor’s algorithm), two numerically different ξ values appear. Earlier documents did not consistently separate these, which may cause confusion.

Binding distinction

Parameter	Value	Domain	Meaning
ξ_{par} (geometric)	$\frac{4}{30000} \approx 1.333 \times 10^{-4}$	3D geometric space	Length hierarchy of the torus; scale levels L_N ; applies in physical position space
ξ_{Higgs} (algebraic)	$\approx 1.038 \times 10^{-5}$	Number space / quantum state space	Algebraic field corrections; Higgs sector; decoherence rates; IBM validation

Justification

$\xi_{\text{par}} = 4/30000$ follows from the torus geometry of physical space (geometric derivation, independent of SI measurements). ξ_{Higgs} follows from the algebraic Higgs field structure and appears in coupling formulas between quantum states (e.g. Bell correlations, T_2 hierarchy).

The two ξ values are not a contradiction, but describe complementary aspects: geometry of space (ξ_{par}) vs. algebra of state space (ξ_{Higgs}). In external communications, it must always be specified which parameter is meant.

7 Clarification 4: L_0 and the Schwarzschild Interpretation

Affected documents

Older versions of Doc. 180–182 (Lagrangian derivation, torus geometry, maximum universe scale).

Erroneous approach

Earlier versions interpreted $L_0 = \xi_{\text{par}} \cdot \ell_P$ as the Schwarzschild radius of a fundamental minimal entity. This interpretation was explicitly rejected in Doc. 182 (revised 2026).

Correct classification

L_0 is exclusively the **geometric fundamental length** of the FFGFT torus:

$$L_0 = \xi_{\text{par}} \cdot \ell_P \approx 2.15 \times 10^{-39} \text{ m.} \quad (190.10)$$

It defines the lowest scale level of the ξ -hierarchy and has *no* Schwarzschild interpretation. The cosmological Hubble radius R_H is the system-specific upper bound of the cosmological sector – not a universal maximum size. Every scale has its own system-specific upper bound.

Note: Schwarzschild as a Consistency Check

The Schwarzschild connection remains valid as an **internal consistency check**: an object with mass $M_0 = \xi_{\text{par}} \cdot m_P/2$ would have Schwarzschild radius exactly L_0 – without any free parameter. If M_0 coincides with a scale level of the ξ -hierarchy, this is a non-trivial self-consistency test. The connection is a *probe, not a derivation*.

8 Clarification 5: Scope and Claim of the Mass Formulas

Context

Several FFGFT documents describe mass derivations for leptons and other particles. The scope of these derivations must be communicated clearly.

Binding formulation

The FFGFT mass formulas (e.g. $m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v$) yield values that agree with experiment to $\sim 1\%$. This is **not a precision calculation**, but a structural proof:

- The mass formulas *follow geometrically* from ξ and v – they are not fitted.
- The remaining $\sim 1\%$ deviations reflect measurement uncertainties and approximations, not structural errors.
- Numerically better calculations than the Standard Model are *not* claimed.

In external texts, avoid: “FFGFT calculates the masses exactly.” Correct: “FFGFT *derives* the mass structure from a single parameter ξ and reproduces the orders of magnitude to $\sim 1\%$.”

9 Clarification 6: Koide Formula Only with Original Values

Binding rule

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \quad (190.11)$$

may only be evaluated with experimentally measured masses (PDG values).

Not permitted: inserting the symbolic FFGFT terms $r_\ell \cdot \xi^{p_\ell} \cdot v$ directly into the formula. The $\sqrt{m_i}$ operation generates cross-terms with mixed ξ -powers that are foreign to the torus structure space. Any such substitution gives $Q \approx 0.6677 \neq 2/3$ and is not a contradiction of the theory but a category error.

Permitted: evaluating FFGFT terms numerically (e.g. $m_\tau = 1783 \text{ MeV}$) and using that result like a measured value in the Koide formula.

Mathematical background: The r_ℓ terms are geometric invariants of their torus modes. The set of admissible pairs (r_i, p_i) is not closed under multiplication: $r_e \cdot r_\mu = 64/15$ belongs to no torus mode. This is the non-closure theorem (Doc. 193). The general analysis of composition limits is in Doc. 192.

Updated Summary of Corrections

No.	Doc.	Incorrect / Un-clear	Correct / Clari-fied
K1	049 (HTML)	$\xi = \lambda_h^2 v^2 / (64 \pi^4 m_h^2)$	$\xi = \lambda_h^2 v^2 / (16 \pi^3 m_h^2)$
K2	116 (Tab.)	$r_\tau = 8/3$	$r_\tau = 25/9$
K3	018 (Ex-plorer)	$a_e = 4\pi / (f \cdot k)$	$a_e = 2\pi^2 / (f \cdot k)$
P1	116 / 158	$\Delta Q < 0.00003\%$ vs. 0.001%	Both correct, different quantities; 0.001% for external use
P2	several	$\hbar$ as postulate	$\hbar$ follows from ξ via $L_0 = \xi \cdot \ell_P$
P3	173-176	Single uniform ξ value	Two parameters: ξ_{par} (geometric) and ξ_{Higgs} (algebraic)
P4	180-182	L_0 as Schwarzschild radius	$L_0 =$ geometric fundamental length; no Schwarzschild interpretation
P5	several	“FFGFT calculates masses exactly”	FFGFT derives mass structure from ξ ; $\sim 1\%$ agreement